TCS3351 CRYPTOGRAPHY AND DATA SECURITY
Tutorial 7 - Mathematical Background II

1. Distinguish between a prime and a composite integer. Specify the prime and composite integers in the range from 1 to 30 by using Sieve of Eratosthenes method.

2. Determine whether the following integers are primes or composite.
   a) 271
   b) 501
   c) 831
   d) 1479
   e) 3513
   f) 6487

3. Solve the following problem using the approximation method of calculating $\Pi(n)$ (upper limit discovered by Gauss and lower limit discovered by Legendre).

   a) Find the number of primes is between 100,000 and 200,000.
   b) Find the number of composite integers is between 100,000 and 200,000.
   c) What is the ratio of primes to composites in the above range.

4. Euler's Totient Function.
   a) State the use of Euler's Totient function and the rules that can be used for finding the function.
   b) Using the appropriate rules, find the value of
      i. $\phi(29)$
      ii. $\phi(47)$
      iii. $\phi(102)$
      iv. $\mathbb{Z}_{37}^*$
      v. $\mathbb{Z}_{80}^*$
      vi. $\mathbb{Z}_{99}^*$

5. For each of the follow problem, solve with the help of either Fermat's Little Theorem or Euler's
   a) $15^{18} \bmod 17$
   b) $5^{15} \bmod 13$
   c) $15^{-1} \bmod 17$
   d) $27^{-1} \bmod 41$
   e) $12^{-1} \bmod 77$
   f) $16^{-1} \bmod 323$

6. Test the following integers whether they are a strong pseudoprime using the Miller-Rabin primality test. (Use at maximum 3 small different bases, such as $a = 2,3,5,7$)
   a) 997
   b) 3801
   c) 8997

7. Solving the following problem using the Chinese Remainder Theorem.
   a) $x \equiv 2 \bmod 7$ and $x \equiv 3 \bmod 9$
   b) Find an integer that has a remainder of 3 when divided by 7 and 13 but is divisible by 12.

8. Solve the following using the square-and-multiply method of fast exponentiation algorithm.
   a) $320^{23} \bmod 461$
   b) $1736^{41} \bmod 2134$
   c) $1234^{12} \bmod 4099$
   d) $491^{37} \bmod 501$